

Ravi Ranjan Kumar

raviranjankumar2003@gmail.com | +91 9546358236 | [LeetCode](#)
[GitHub](#) | [LinkedIn](#)

Summary

Final-year Computer Science student with hands-on experience building 4+ end-to-end AI systems, including a face recognition attendance platform, an LLM-based resume screening tool, and a medical image classifier. Proficient in Python, Java, TensorFlow, FastAPI, and React. Strong foundations in Machine Learning, Deep Learning, and full-stack development with a focus on deploying real-world applications.

Education

Chandigarh University

B.E. in Computer Science and Engineering

Apr 2022 - Jul 2026

Projects

Smart AI Attendance System [GitHub](#)

Python, Streamlit, Supabase, OpenCV

AI-powered biometric attendance management platform

- Engineered a face recognition attendance platform, reducing manual attendance effort by over 80%.
- Designed teacher and student dashboards for subject creation, enrollment, and real-time attendance tracking.
- Integrated Supabase backend for secure cloud storage and live attendance records.
- Deployed biometric verification workflows supporting multiple concurrent classrooms.

AI-Powered Resume ATS & Candidate Intelligence System [GitHub](#)

Python, FastAPI, React, FAISS, RAG

LLM-based resume screening and ranking system

- Built a full-stack Resume ATS platform automating resume parsing, scoring, and candidate ranking.
- Implemented semantic skill matching using text embeddings and FAISS vector search for high-accuracy shortlisting.
- Engineered batch resume processing pipeline, cutting manual screening time significantly.
- Delivered ranked candidate reports via React frontend, improving recruiter decision speed.

ResearchMind - Multi-Agent AI Research System [GitHub](#)

Python, LangChain, Streamlit, Tavily, Mistral AI

Agentic AI-powered research automation platform

- Built a multi-agent AI system with Search, Reader, Writer, and Critic agents for automated research generation.
- Integrated Tavily Search and web scraping tools to gather and analyze information from online sources.
- Generated structured research reports with automated evaluation, scoring, and feedback mechanisms.
- Developed a Streamlit interface for real-time research execution and report downloads.

Pneumonia Detection Using Deep Learning [GitHub](#)

Python, TensorFlow, ResNet-50, OpenCV

Medical image classification system

- Built a pneumonia detection classifier using ResNet-50 transfer learning on 5,856 labeled chest X-ray images.
- Applied image augmentation and preprocessing techniques to improve model generalization and accuracy.
- Achieved robust classification performance suitable for supporting early clinical diagnosis workflows.
- Designed the pipeline to be extensible for other medical imaging classification tasks.

Technical Skills

Programming Languages: Python, Java, SQL

Web / Backend: FastAPI, React, Streamlit, REST APIs

Databases / Cloud: MySQL, PostgreSQL, Supabase

AI / ML: Machine Learning, Deep Learning, LLMs, LangChain, RAG, Computer Vision

Libraries / Tools: TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, OpenCV, Matplotlib

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems

Certifications

- Cloud Computing - NPTEL, IIT Kharagpur (Silver Medal)
- Computer Organization and Architecture - NPTEL, IIT Guwahati
- IoT Architectures and Protocols - NPTEL, NITTR Chandigarh